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CREATE TABLE customers (
    customer_id INT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    city VARCHAR(50) NOT NULL,
    state VARCHAR(50) NOT NULL,
    join_date DATE NOT NULL,
    is_premium BOOLEAN DEFAULT FALSE
); -- Index for filtering by city/state
CREATE INDEX idx_customers_city ON customers(city);
CREATE INDEX idx_customers_state ON customers(state);

CREATE TABLE products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100) NOT NULL,
    category VARCHAR(50) NOT NULL,
    brand VARCHAR(50) NOT NULL,
    unit_price DECIMAL(10,2) NOT NULL CHECK (unit_price > 0),
    stock_qty INT NOT NULL DEFAULT 0 CHECK (stock_qty >= 0)
); -- Index for filtering by category
CREATE INDEX idx_products_category ON products(category);

CREATE TABLE orders (
    order_id INT PRIMARY KEY,
    customer_id INT NOT NULL,
    order_date DATE NOT NULL,
    status VARCHAR(20) NOT NULL DEFAULT 'Pending'
        CHECK (status IN ('Pending', 'Shipped', 'Delivered', 'Cancelled')),
    total_amount DECIMAL(12,2) NOT NULL CHECK (total_amount >= 0),
    FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
); -- Index for date-based filtering and sorting
CREATE INDEX idx_orders_date ON orders(order_date);
CREATE INDEX idx_orders_status ON orders(status);

CREATE TABLE order_items (
    item_id INT PRIMARY KEY,

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order_id INT NOT NULL,
product_id INT NOT NULL,
quantity INT NOT NULL CHECK (quantity > 0),
unit_price DECIMAL(10,2) NOT NULL CHECK (unit_price > 0),
discount_pct DECIMAL(5,2) DEFAULT 0 CHECK (discount_pct BETWEEN 0 AND 100),
FOREIGN KEY (order_id) REFERENCES orders(order_id),
FOREIGN KEY (product_id) REFERENCES products(product_id)
);

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INSERT INTO customers VALUES

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(101, 'Aarav', 'Sharma', 'aarav.s@email.com', 'Mumbai', 'Maharashtra', '2024-01-15', TRUE),
(102, 'Priya', 'Patel', 'priya.p@email.com', 'Ahmedabad', 'Gujarat', '2024-02-20', FALSE),
(103, 'Rohan', 'Gupta', 'rohan.g@email.com', 'Delhi', 'Delhi', '2024-03-10', TRUE),
(104, 'Sneha', 'Reddy', 'sneha.r@email.com', 'Hyderabad', 'Telangana', '2024-04-05', FALSE),
(105, 'Vikram', 'Singh', 'vikram.s@email.com', 'Jaipur', 'Rajasthan', '2024-05-12', TRUE),
(106, 'Ananya', 'Iyer', 'ananya.i@email.com', 'Chennai', 'Tamil Nadu', '2024-06-18', FALSE),
(107, 'Karan', 'Mehta', 'karan.m@email.com', 'Pune', 'Maharashtra', '2024-07-22', TRUE),
(108, 'Divya', 'Nair', 'divya.n@email.com', 'Kochi', 'Kerala', '2024-08-30', FALSE);

```

select * from customers;

55								
56	INSERT INTO customers VALUES							
57	(101, 'Aarav', 'Sharma', 'aarav.s@email.com', 'Mumbai', 'Maharashtra', '2024-01-15', TRUE),							
58	(102, 'Priya', 'Patel', 'priya.p@email.com', 'Ahmedabad', 'Gujarat', '2024-02-20', FALSE),							
59	(103, 'Rohan', 'Gupta', 'rohan.g@email.com', 'Delhi', 'Delhi', '2024-03-10', TRUE),							
60	(104, 'Sneha', 'Reddy', 'sneha.r@email.com', 'Hyderabad', 'Telangana', '2024-04-05', FALSE),							
61	(105, 'Vikram', 'Singh', 'vikram.s@email.com', 'Jaipur', 'Rajasthan', '2024-05-12', TRUE),							
62	(106, 'Ananya', 'Iyer', 'ananya.i@email.com', 'Chennai', 'Tamil Nadu', '2024-06-18', FALSE),							
63	(107, 'Karan', 'Mehta', 'karan.m@email.com', 'Pune', 'Maharashtra', '2024-07-22', TRUE),							
64	(108, 'Divya', 'Nair', 'divya.n@email.com', 'Kochi', 'Kerala', '2024-08-30', FALSE);							
65								
66	select * from customers;							
67								
68	INSERT INTO products VALUES							
	customer_id	first_name	last_name	email	city	state	join_date	is_premium
1	101	Aarav	Sharma	aarav.s@email.com	Mumbai	Maharashtra	2024-01-15	1
2	102	Priya	Patel	priya.p@email.com	Ahmedabad	Gujarat	2024-02-20	0
3	103	Rohan	Gupta	rohan.g@email.com	Delhi	Delhi	2024-03-10	1
4	104	Sneha	Reddy	sneha.r@email.com	Hyderabad	Telangana	2024-04-05	0
5	105	Vikram	Singh	vikram.s@email.com	Jaipur	Rajasthan	2024-05-12	1
6	106	Ananya	Iyer	ananya.i@email.com	Chennai	Tamil Nadu	2024-06-18	0
7	107	Karan	Mehta	karan.m@email.com	Pune	Maharashtra	2024-07-22	1
8	108	Divya	Nair	divya.n@email.com	Kochi	Kerala	2024-08-30	0

INSERT INTO products VALUES

```
(201, 'Wireless Earbuds', 'Electronics', 'BoAt', 1499.00, 250),
(202, 'Cotton T-Shirt', 'Clothing', 'Levis', 799.00, 500),
(203, 'Smart Watch', 'Electronics', 'Noise', 2999.00, 150),
(204, 'Running Shoes', 'Clothing', 'Nike', 4599.00, 120),
(205, 'Bluetooth Speaker', 'Electronics', 'JBL', 3499.00, 200),
(206, 'Bedsheet Set', 'Home', 'Spaces', 1299.00, 300),
(207, 'Laptop Stand', 'Electronics', 'AmazonBasics', 899.00, 180),
(208, 'Cushion Covers (Set)', 'Home', 'HomeCenter', 599.00, 400);
```

```
select * from products ;
```

```
66 select * from customers;
67
68 INSERT INTO products VALUES
69 (201, 'Wireless Earbuds', 'Electronics', 'BoAt', 1499.00, 250),
70 (202, 'Cotton T-Shirt', 'Clothing', 'Levis', 799.00, 500),
71 (203, 'Smart Watch', 'Electronics', 'Noise', 2999.00, 150),
72 (204, 'Running Shoes', 'Clothing', 'Nike', 4599.00, 120),
73 (205, 'Bluetooth Speaker', 'Electronics', 'JBL', 3499.00, 200),
74 (206, 'Bedsheet Set', 'Home', 'Spaces', 1299.00, 300),
75 (207, 'Laptop Stand', 'Electronics', 'AmazonBasics', 899.00, 180),
76 (208, 'Cushion Covers (Set)', 'Home', 'HomeCenter', 599.00, 400);
77
78 select * from products ;
79
```

	product_id	product_name	category	brand	unit_price	stock_qty
1	201	Wireless Earbuds	Electronics	BoAt	1499	250
2	202	Cotton T-Shirt	Clothing	Levis	799	500
3	203	Smart Watch	Electronics	Noise	2999	150
4	204	Running Shoes	Clothing	Nike	4599	120
5	205	Bluetooth Speaker	Electronics	JBL	3499	200
6	206	Bedsheet Set	Home	Spaces	1299	300
7	207	Laptop Stand	Electronics	AmazonBasics	899	180
8	208	Cushion Covers (Set)	Home	HomeCenter	599	400

```
INSERT INTO orders VALUES
```

```
(1001, 101, '2024-08-01', 'Delivered', 4498.00),
(1002, 102, '2024-08-03', 'Delivered', 799.00),
(1003, 103, '2024-08-05', 'Shipped', 7498.00),
(1004, 101, '2024-08-10', 'Delivered', 3499.00),
(1005, 104, '2024-08-12', 'Cancelled', 2999.00),
(1006, 105, '2024-08-15', 'Delivered', 5898.00),
(1007, 106, '2024-08-18', 'Pending', 1299.00),
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(1008, 103, '2024-08-20', 'Delivered', 899.00),
(1009, 107, '2024-08-25', 'Shipped', 6098.00),
(1010, 108, '2024-08-28', 'Delivered', 1598.00);
```

```
select * from orders ;
```

81	INSERT INTO orders VALUES				
82	(1001, 101, '2024-08-01', 'Delivered', 4498.00),				
83	(1002, 102, '2024-08-03', 'Delivered', 799.00),				
84	(1003, 103, '2024-08-05', 'Shipped', 7498.00),				
85	(1004, 101, '2024-08-10', 'Delivered', 3499.00),				
86	(1005, 104, '2024-08-12', 'Cancelled', 2999.00),				
87	(1006, 105, '2024-08-15', 'Delivered', 5898.00),				
88	(1007, 106, '2024-08-18', 'Pending', 1299.00),				
89	(1008, 103, '2024-08-20', 'Delivered', 899.00),				
90	(1009, 107, '2024-08-25', 'Shipped', 6098.00),				
91	(1010, 108, '2024-08-28', 'Delivered', 1598.00);				
92					
93	select * from orders ;				
94					

	order_id	customer_id	order_date	status	total_amount
1	1001	101	2024-08-01	Delivered	4498
2	1002	102	2024-08-03	Delivered	799
3	1003	103	2024-08-05	Shipped	7498
4	1004	101	2024-08-10	Delivered	3499
5	1005	104	2024-08-12	Cancelled	2999
6	1006	105	2024-08-15	Delivered	5898
7	1007	106	2024-08-18	Pending	1299
8	1008	103	2024-08-20	Delivered	899
9	1009	107	2024-08-25	Shipped	6098
10	1010	108	2024-08-28	Delivered	1598

```
INSERT INTO order_items VALUES
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```
(5001, 1001, 201, 2, 1499.00, 0),
(5002, 1001, 207, 1, 899.00, 10),
(5003, 1002, 202, 1, 799.00, 0),
(5004, 1003, 203, 1, 2999.00, 0),
(5005, 1003, 204, 1, 4599.00, 5),
(5006, 1004, 205, 1, 3499.00, 0),
(5007, 1005, 203, 1, 2999.00, 0),
(5008, 1006, 201, 1, 1499.00, 10),
(5009, 1006, 204, 1, 4599.00, 5),
(5010, 1007, 206, 1, 1299.00, 0),
(5011, 1008, 207, 1, 899.00, 0),
(5012, 1009, 205, 1, 3499.00, 0),
(5013, 1009, 208, 2, 599.00, 15),
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(5014, 1010, 206, 1, 1299.00, 0),

(5015, 1010, 208, 1, 599.00, 0);

select * from order_items ;

```
SQL 1
97 (5002, 1001, 207, 1, 899.00, 10),
98 (5003, 1002, 202, 1, 799.00, 0),
99 (5004, 1003, 203, 1, 2999.00, 0),
100 (5005, 1003, 204, 1, 4599.00, 5),
101 (5006, 1004, 205, 1, 3499.00, 0),
102 (5007, 1005, 203, 1, 2999.00, 0),
103 (5008, 1006, 201, 1, 1499.00, 10),
104 (5009, 1006, 204, 1, 4599.00, 5),
105 (5010, 1007, 206, 1, 1299.00, 0),
106 (5011, 1008, 207, 1, 899.00, 0),
107 (5012, 1009, 205, 1, 3499.00, 0),
108 (5013, 1009, 208, 2, 599.00, 15),
109 (5014, 1010, 206, 1, 1299.00, 0),
110 (5015, 1010, 208, 1, 599.00, 0);
111
112 select * from order_items ;
113
```

	item_id	order_id	product_id	quantity	unit_price	discount_pct
1	5001	1001	201	2	1499	0
2	5002	1001	207	1	899	10
3	5003	1002	202	1	799	0
4	5004	1003	203	1	2999	0
5	5005	1003	204	1	4599	5
6	5006	1004	205	1	3499	0
7	5007	1005	203	1	2999	0
8	5008	1006	201	1	1499	10
9	5009	1006	204	1	4599	5
10	5010	1007	206	1	1299	0